Engine Model(s): PT6A

74-10-00

IGNITION EXCITER - DESCRIPTION AND OPERATION

1. Description and Operation

The ignition exciter is a sealed unit containing electronic components encased in an epoxy resin. The unit is energized only during the engine starting sequence to initiate combustion in the combustion chamber. The exciter transforms the DC input to a pulsed high voltage output through solid-state circuitry, a transformer and diodes.

When the unit is energized, a capacitor on the high voltage side of the output transformer is progressively charged, until the energy stored, approximately four joules, is sufficient to ionize a spark gap in the unit and discharge the capacitor across the two spark igniters through a dividing and step-up transformer network. The network is designed so that if one igniter is open or shorted, the remaining igniter will continue to function. The network also enables the capacitor to discharge automatically in the event of either or both igniters becoming inoperative, or input voltage being switched off.

Generally, the ignition exciter is engine-bracket mounted in either a vertical or horizontal attitude and some brackets may have vibration absorption mounts installed (Ref. SB1422), depending on engine model and airframe application. However, on certain airframe applications, the exciter may be remote-mounted on the airframe.

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